ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

INVERTIS UNIVERSITY BAREILLY

EVEN SEMESTER EXAMINATION - 2024-25

B.Tech.(CSE) (AI)

Semester: II

SUBJECT CODE: BCSAI202

PROBLEM SOLVING IN C

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. (10 Marks)

Q. 1. Answer the following questions: (1 × 5 = 5 Marks)

(i) What is a dangling pointer in C?

(ii) Explain the significance of fread() in file handling.

(iii) How is calloc() different from malloc() in memory allocation?

(iv) What are the advantages of using macros in C programming?

(v) Differentiate between a process and a thread in C.

Q. 2. Answer the following questions: (1 × 5 = 5 Marks)

(i) What is the advantage of using a structure over an array?

(ii) Explain the use of fseek() and rewind() in file handling.

(iii) What is external fragmentation in memory management?

(iv) How do standard header files help in C programming?

(v) What is the use of pthread_create() in multithreading?

SECTION – B

NOTE:Attempt all questions. (5×2×2 = 20 Marks)

Q.3.(i)(a) How does a function pointer work? Explain with an example.

OR

(b) Describe a circular linked list and how it differs from a doubly linked list.

(ii)(a) Explain different file handling modes in C.

OR

(b) How do fprintf() and fread() functions work? Illustrate with an example.

Q.4.(i)(a) What is a critical section in multithreading? Explain its significance.

OR

(b) Explain the concept of starvation and deadlock in multithreading.

(ii)(a) Describe the properties and working of a B-tree with an example.

OR

(b) Explain separate chaining and double hashing methods in hashing.

SECTION – C

NOTE: Attempt all questions. (10 × 2 = 20 Marks)

Q.5.(a) Explain how macros are expanded at the preprocessing stage with an example.

OR

(b) What is inter-process communication (IPC) in socket programming? Explain with an example.

Q.6.(a) Write a program to implement a circular linked list in C with insertion and deletion operations.

OR

(b) Discuss the role of dynamic memory allocation functions (malloc, calloc, realloc, free) with examples.